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Which one of the following statements is NOT CORRECT?

- ① A well-conditioned matrix has a condition number close to 1
- ② An ill-conditioned matrix has a large condition number
- ③ The inverse of a well-conditioned matrix can be computed accurately
- ④ A non-invertible matrix has a condition number close to 1

Theoretical Solution

The **condition number** of an invertible matrix $\mathbf{A}$, denoted as $\kappa(\mathbf{A})$, is defined as:

$$\kappa(\mathbf{A}) = \|\mathbf{A}\| \left\| \mathbf{A}^{-1} \right\| \quad (1)$$

where,

$$\|\mathbf{A}\| = \text{norm of matrix } \mathbf{A}$$

It measures how much the output of a linear system $\mathbf{Ax} = b$ can change for a small change in the input data.

Theoretical Solution

In linear algebra, a **matrix norm** is a function that assigns a strictly positive length or size to every non-zero matrix. Conceptually, it is an extension of the idea of a vector norm (which measures the length of a vector) to matrices.

A function $\|\mathbf{A}\|$ is defined as a matrix norm if it satisfies the following properties for any matrices $\mathbf{A}$ and $\mathbf{B}$ and any scalar c :

- ① **Non-negativity:** $\|\mathbf{A}\| \geq 0$, and $\|\mathbf{A}\| = 0$ if and only if $\mathbf{A}$ is the zero matrix.
- ② **Absolute Homogeneity:** $\|c\mathbf{A}\| = |c| \cdot \|\mathbf{A}\|$.
- ③ **Triangle Inequality:** $\|\mathbf{A} + \mathbf{B}\| \leq \|\mathbf{A}\| + \|\mathbf{B}\|$.
- ④ **Submultiplicativity:** $\|\mathbf{AB}\| \leq \|\mathbf{A}\| \cdot \|\mathbf{B}\|$. (This property is specific to matrix norms).

Theoretical Solution

There are several different ways to define a matrix norm, with the most common types falling into two categories: induced norms and entry-wise norms.

These norms are defined in terms of how a matrix acts on vectors. An induced norm measures the maximum "stretching factor" that the matrix applies to any vector. It is defined as:

$$\|\mathbf{A}\|_p = \sup_{\mathbf{x} \neq 0} \frac{\|\mathbf{Ax}\|_p}{\|\mathbf{x}\|_p} \quad (2)$$

The most common induced norms are:

- **L1-Norm (Maximum Absolute Column Sum):**

$$\|\mathbf{A}\|_1 = \max_j \sum_{i=1}^m |a_{ij}| \quad (3)$$

Theoretical Solution

- **L-infinity Norm (Maximum Absolute Row Sum):**

$$\|\mathbf{A}\|_{\infty} = \max_i \sum_{j=1}^n |a_{ij}| \quad (4)$$

- **L2-Norm (Spectral Norm):**

$$\|\mathbf{A}\|_2 = \sqrt{\lambda_{\max}(\mathbf{A}^* \mathbf{A})} = \sigma_{\max}(\mathbf{A}) \quad (5)$$

where $\sigma_{\max}$ is the largest singular value of $\mathbf{A}$ and $\mathbf{A}^*$ is the conjugate transpose of $\mathbf{A}$.

These norms treat the matrix as a long vector of its $m \times n$ elements.

- **Frobenius Norm:**

$$\|\mathbf{A}\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n |a_{ij}|^2} \quad (6)$$

This is analogous to the standard Euclidean norm for vectors.

- 1) **A well-conditioned matrix has a condition number close to 1:** This is **CORRECT**. The smallest possible condition number is 1 (for example, the identity matrix). A value close to 1 indicates that the matrix is numerically stable.
- 2) **An ill-conditioned matrix has a large condition number:** This is **CORRECT**. This is the definition of an ill-conditioned matrix. A large condition number means that small errors in the input can lead to very large errors in the solution.

- 3) **The inverse of a well-conditioned matrix can be computed accurately:** This is **CORRECT**. Because a well-conditioned matrix is not sensitive to small perturbations, numerical algorithms used to compute its inverse will produce results with high accuracy.
- 4) **A non-invertible matrix has a condition number close to 1:** This is **NOT CORRECT**. A non-invertible (or singular) matrix does not have an inverse $\mathbf{A}^{-1}$. For the purposes of conditioning, its inverse is considered to have an infinite norm. Therefore, the condition number of a non-invertible matrix is defined to be **infinity** (∞). An infinite condition number is the largest possible value, representing the worst possible conditioning, not the best (which is close to 1).